

Weak Roman Domination on Graphs

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Abstract

*Weak Roman domination problem is a variation of domination problem, and it has practical applications in handling resource sharing. Let $f: V \rightarrow \{0, 1, 2\}$ be a function of $G = (V, E)$ such that $V_i = \{u \mid f(u) = i\}$ for $0 \leq i \leq 2$. A vertex u is considered to be *unsecured* if $u \in V_0$ and *secured* otherwise. A vertex $u \in V_0$ is said to be *undefended* if it is not adjacent to any vertex in V_1 or V_2 . Function f is said to be *safe* if there is no undefended vertex. For each $u \in V_0$, a move from a neighborhood vertex $v \in V_1 \cup V_2$ to u respect to f is defined as the function $f': V \rightarrow \{0, 1, 2\}$ such that $f'(u) = 1$, $f'(v) = f(v) - 1$ and $f'(w) = f(w)$ for $w \in V - \{u, v\}$. If every vertex in V_0 , there is a safe move f' respectively to the safe function f then f is called a *weak Roman dominating function*. The cost of a weak Roman dominating function is defined as $w(f) = \sum_{v \in V} f(v) = 2|V_2| + |V_1|$. The weak Roman domination number of G , denote as $\gamma_w(G)$ is minimum cost among all weak Roman dominating functions of G . This paper established the weak Roman domination number of corona of any two graphs and generalized web graphs.*

1 Introduction

The Roman domination problem is a variation of domination problem suggested by [6]. In the third century, Roman Empire occupied most area and it was very powerful with enough legions throughout the empire. Each town belongs to the Rome was prosperous and was able to keep at least one legion to protect themselves from invasion. By 4th century, the Empire was weakened and some towns could not afford to keep a legion of their own but had to be protected by others. In order not to sacrifice itself while saving others, Emperor Constantine the Great issued a new degree that only the town with two legions could dispatch one to protect the adjacent towns. The problem can be modeled

by a graph. Let each vertex denotes a town and it is considered to be *unsecured* if there is no legion stationed in it and *secured* otherwise. A *Roman dominating function (RDF)* of $G(V, E)$ is a function $f: V \rightarrow \{0, 1, 2\}$ such that $V_i = \{u \mid f(u) = i\}$ for $0 \leq i \leq 2$, and each $u \in V_0$ is adjacent to a vertex in V_2 .

In 2003, Henning and Hedetniemi proposed a new strategy which required fewer legions to protect the Roman Empire [3, 4]. They defined a vertex $u \in V_0$ to be *undefended* if $N(u) \subseteq V_0$ where $N(u) = \{v \mid u \in E(G)\}$. A function f is said to be *safe* if there is no undefended vertex. For each $u \in V_0$, we define a *move* from an adjacent vertex $v \in V_1 \cup V_2$ to u respect to f , to be a function $f': V \rightarrow \{0, 1, 2\}$ such that $f'(u) = 1$, $f'(v) = f(v) - 1$ and $f'(w) = f(w)$ for $w \in V - \{u, v\}$. If both function f and f' are safe, then f is called a *weak Roman dominating function (WRDF)* of the graph G with cost (or weight) $w(f) = \sum_{v \in V} f(v) = 2|V_2| + |V_1|$. The *weak Roman domination number* of G , denoted as $\gamma_w(G)$, is the minimum cost among all WRDFs of G . f is an optimal WRDF of G if and only if $w(f) = \gamma_w(G)$.

This paper established the weak Roman domination number of corona graphs, and provided algorithms to station the minimum number of legions to protect some generalized web graphs.

2 Weak Roman Domination of Corona of Two Graphs

The *corona* of two graphs $G = (V_1, E_1)$ and $H = (V_2, E_2)$, denoted as $G \wedge H$, has vertex set $V(G \wedge H) = V \cup \{n \text{ distinct copies of } V_2 \text{ denoted as } V_{2,1}, V_{2,2}, \dots, V_{2,n-1}, V_{2,n}\}$ and edge set $E(G \wedge H) = E \cup \{n \text{ distinct copies of } E_2 \text{ de-$

noted as $E_{2,1}, E_{2,2}, \dots, E_{2,n-1}, E_{2,n}\} \cup \{(u_i, v) : u_i \in V, v \in V_{2,i}\}$. Figure 1 shows an example of $C_6 \wedge P_2$.

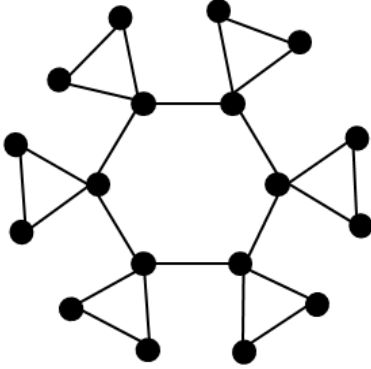


Figure 1: Corona of two graphs $C_6 \wedge P_2$.

For $G \wedge H$, let $V(G) = \{v_1, v_2, \dots, v_n\}$ and $V(H_i) = \{u_{i,1}, u_{i,2}, \dots, u_{i,m-1}, u_{i,m}\}$ for $1 \leq i \leq n$.

Theorem 1 Let G be a graph of n vertices. The weak Roman domination number of $G \wedge K_m$ is $\gamma_w(G \wedge K_m) = n$.

Proof. Define $f : V(G \wedge K_m) \rightarrow \{0, 1, 2\}$ such that $f(v_i) = 1$ and $f(u_{i,j}) = 0$ for $1 \leq i \leq n$ and $1 \leq j \leq m$. Then $V_0 = \{u_{i,j} \mid 1 \leq i \leq n, 1 \leq j \leq m\}$, $V_1 = \{v_i \mid 1 \leq i \leq n\}$ and $V_2 = \emptyset$. Since each vertex in V_0 is adjacent to a vertex in V_1 , f is a safe function. For each $u_{i,j} \in V_0$, consider a move f' such that $f'(u_{i,j}) = 1$ and $f'(v_i) = f(v_i) - 1 = 0$, $f'(w) = f(w)$ for all $w \in V(G \wedge K_m) - \{u_{i,j}, v_i\}$. Then $V'_0 = V_0 - u_{i,j} + v_i$, $V'_2 = \emptyset$ and $V'_1 = V_1 - v_i + u_{i,j}$. Since $\{u_{i,k}, u_{i,j}\} \in E(G \wedge K_m)$ for each $1 \leq k \neq j \leq m$, $N(v_i) \cap V_0 \subseteq N(u_{i,j})$, f' is a safe function. Hence, f is a WRDF of $G \wedge K_m$, which implies that $\gamma_w(G \wedge K_m) \leq n$.

To see that $\gamma_w(G \wedge K_m) \geq n$, suppose to the contrary that $\gamma_w(G \wedge K_m) \leq n-1$. Let g be an optimal WRDF of $G \wedge K_m$ with $w(g) = \gamma_w(G \wedge K_m) \leq n-1$. Then there exists some $1 \leq i \leq n$ such that $g(v_i) + \sum_{j=1}^m g(u_{i,j}) = 0$

which implies $N(u_{i,j}) \subseteq V_0$, then $u_{i,j}$ is undefended. That is, g is not a safe function, which contradict to g being a WRDF of $G \wedge K_m$. Hence, $\gamma_w(G \wedge H) = n$. $\square$

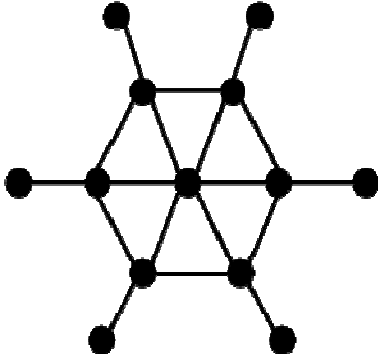
Theorem 2 Let G and H be two graphs with n and m vertices respectively and $H \neq K_m$, then $\gamma_w(G \wedge H) = 2n$.

Proof. Define $f : V(G \wedge H) \rightarrow \{0, 1, 2\}$ such that $f(v_i) = 2$ and $f(u_{i,j}) = 0$ for $1 \leq i \leq n$ and $1 \leq j \leq m$. Then $V_0 = \{u_{i,j} \mid 1 \leq i \leq n, 1 \leq j \leq m\}$, $V_2 = \{v_i \mid 1 \leq i \leq n\}$ and $V_1 = \emptyset$. Since each vertex in V_0 is adjacent to a vertex in V_2 , f is a safe function. For each $u_{i,j} \in V_0$, consider a move f' such that $f'(u_{i,j}) = 1$ and $f'(v_i) = f(v_i) - 1 = 1$, $f'(w) = f(w)$ for each $w \in V(G \wedge H) - \{u_{i,j}, v_i\}$. Then $V'_0 = V_0 - u_{i,j}$, $V'_1 = \{u_{i,j}, v_i\}$, $V'_2 = V_2 - v_i$. Since f' does not create any new unsecured vertex, f' is safe. Hence, f is a WRDF of $G \wedge H$, which implies that $\gamma_w(G \wedge H) \leq 2n$.

To see that $\gamma_w(G \wedge H) \geq 2n$, suppose to the contrary that $\gamma_w(G \wedge H) \leq 2n-1$. Let g be a WRDF of $G \wedge H$ with $w(g) = \gamma_w(G \wedge H) \leq 2n-1$. Then there exists some $1 \leq i \leq n$ such that $g(v_i) + \sum_{j=1}^m g(u_{i,j}) < 2$. Since $H_i \neq K_m$, there are at least two vertices $u_{i,j}$ and $u_{i,k}$ such that $(u_{i,j}, u_{i,k}) \notin E(G \wedge H)$ for $k \neq j$. Then consider that there is a move to $u_{i,j}$, $u_{i,k}$ becomes undefended which contradict to g is a safe WRDF of $G \wedge H$. Hence, $\gamma_w(G \wedge H) = 2n$. $\square$

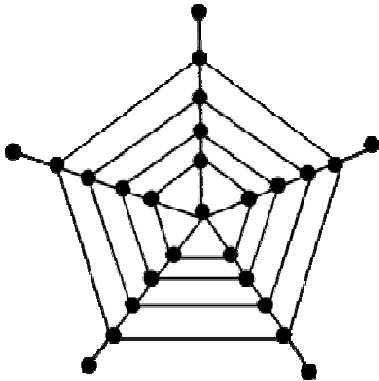
3 Weak Roman Domination on Generalized Web Graphs

A *helm* graph H_n is a graph obtained from a *wheel* by attaching a pendant edge at each vertex of the n -cycle [2] (see Figure 2 as an example of H_6).

Figure 2: Helm graph H_6 .

A web graph W_n , which is produce by joining a pendant vertex of a helm to form a cycle and then adding a single pendant edge to each vertex of this outer cycle, is proposed in 2003 [5].

Yang [7] extended the notion of a web by iterating the process of adding pendant edges, joining them to form a cycle, and adding pendant vertices to the new cycle. They used $W_{2,n}$ to represent the web W_n and $W_{t,n}$ to represent a generalized web graph with t n -cycles as shown in Figure 3.

Figure 3: Generalized web graphs $W_{4,5}$.

In this paper, let v_0 denotes the central vertex of $W_{t,n}$. Let $C_n^1 = v_{1,1}, v_{1,2}, \dots, v_{1,n}, v_{1,1}$ represent the inner cycle, $C_n^i = v_{i,1}, v_{i,2}, \dots, v_{i,n}, v_{i,1}$ represent the middle cycles for $2 \leq i \leq t-1$ and $C_n^t = v_{t,1}, v_{t,2}, \dots, v_{t,n}, v_{t,1}$ represent the outer cycle of $W_{t,n}$. Let u_k denote the pendant vertices of $v_{t,k}$ in $W_{t,n}$ for $1 \leq k \leq n$.

Proposition 1 Let f be a WRDF of $W_{t,n}$ for $t \geq 3$ with $w(f) = \gamma_w(W_{t,n})$. Then $f(v_{t,k}) + f(u_k) = 1$ for $1 \leq k \leq n$.

Proposition 2 Let f be a safe function of G

with $f(v) = 2$ for some $v \in V(G)$. Then there is a safe move from v to any vertex $u \in N(v)$.

Proposition 3 Let f be a WRDF of G . Suppose u is a pendant vertex of v , if $f(v) = 1$ and $f(u) = 0$, then the only safe move from u is to v .

Proposition 4 Let f be a safe function of $W_{t,n}$. Suppose $\{u, v\} \subseteq V_0$, $(u, v) \notin E(W_{t,n})$ and $(N(u) \cup N(v)) \cap (V_1 \cup V_2) = \{w\}$, if $f(w) < 2$, then f is not a WRDF.

Proposition 5 Let f be a safe function of $W_{t,n}$ and $f(v_{i,j}) = 1$. If $\{v_{i+1,j-1}, v_{i-1,j+1}\} \subseteq V_1 \cup V_2$, then there are safe moves from $v_{i,j}$ to $N(v_{i,j})$. Furthermore, If $N(v_{i-2,j+1}) \cap (V_1 \cup V_2) \neq \emptyset$ then there is a safe move from $v_{i-1,j+1}$ to $v_{i-1,j+2}$; If $N(v_{i-2,j-1}) \cap (V_1 \cup V_2) \neq \emptyset$ then there is a safe move from $v_{i-1,j-1}$ to $v_{i-1,j-2}$.

Proposition 6 Let f be a safe function of $W_{t,n}$ and $f(v_{i,j}) = 1$. If $\{v_{i-1,j-1}, v_{i-1,j+1}\} \subseteq V_1 \cup V_2$, then there are safe move from $v_{i,j}$ to $v_{i+1,j}$; If $\{v_{i+1,j-1}, v_{i+1,j+1}\} \subseteq V_1 \cup V_2$, then there are safe move from $v_{i,j}$ to $v_{i-1,j}$.

Theorem 3 Let G be a helm H_n or a web W_n , the weak Roman domination number of G is $\gamma_w(G) = n+1$.

Proof. First we show that $\gamma_w(H_n) = n+1$. Define $f : V(H_n) \rightarrow \{0, 1, 2\}$ as $f(v_{1,k}) = 1$, $f(u_k) = 0$ for $1 \leq k \leq n$ and $f(v_0) = 1$. Then $V_0 = \{u_k\}$, $V_1 = \{v_{1,k}, v_0\}$ for $1 \leq k \leq n$ and $V_2 = \emptyset$ and $w(f) = n+1$. Since each vertex in V_0 is adjacent to a vertex in V_1 , f is a safe function with cost $n+1$. For each $u_k \in V_0$, consider a move f' such that $f'(u_k) = 1$, $f'(v_{1,k}) = f(v_{1,k}) - 1 = 0$ and $f'(w) = f(w)$ for each $w \in V(H_n) - \{u_k, v_{1,k}\}$. Then, $V_0' = V_0 - u_k + v_{1,k}$, $V_1' = V_1 - v_{1,k} + u_k$ and $V_2' = \emptyset$.

Notice that, $N(v_{1,k}) = \{v_{1,k+1}, v_{1,k-1}, v_0, u_k\} \subseteq V_1'$ for $1 \leq k \leq n$. Hence, f' is safe, which implies f is a WRDF of H_n . Therefore, $\gamma_w(H_n) \leq n+1$.

To see that $\gamma_w(H_n) \geq n+1$, suppose to the contrary that $\gamma_w(H_n) \leq n$. Let g be a WRDF of H_n with $w(g) = \gamma_w(H_n) \leq n$. By proposition 1, $g(u_k) + g(v_{1,k}) = 1$ then we have $g(v_0) = 0$. Any move from $v_{1,k}$ to v_0 will result u_k to be undefended which contradict to g is a WRDF of H_n . Therefore, $\gamma_w(H_n) = n+1$.

Next we show that $\gamma_w(W_n) = n+1$. Define $f: V(G) \rightarrow \{0,1,2\}$ as $f(v_{2,k}) = 1$, $f(v_{1,k}) = 0$, $f(u_k) = 0$ for $1 \leq k \leq n$ and $f(v_0) = 1$, then $V_0 = \{v_{1,k}, u_k\}$ for $1 \leq k \leq n$, $V_1 = \{v_{2,k}, v_0\}$ for $1 \leq k \leq n$, $V_2 = \emptyset$ and $w(f) = n+1$. Since each vertex in V_0 is adjacent to a vertex in V_1 , f is a safe function. For each $u_k \in V_0$, consider a move f' such that $f'(u_k) = 1$, $f'(v_{2,k}) = f(v_{2,k}) - 1 = 0$ and $f'(w) = f(w)$ for each $w \in V(G) - \{u_k, v_{2,k}\}$. Then, $V_0' = V_0 - u_k + v_{2,k}$, $V_2' = \emptyset$ and $V_1' = V_1 - v_{2,k} + u_k$. Since $N(v_{2,k}) = \{v_{2,k+1}, v_{2,k-1}, v_{1,k}, u_k\}$, $\{v_{2,k+1}, v_{2,k-1}, u_k\} \subseteq V_1'$ and $N(v_{1,k}) \cap V_1 = \{v_0\}$, f' is safe. For each $v_{1,k} \in V_0$ ($1 \leq k \leq n$), consider a move g such that $g(v_{1,k}) = 1$, $g(v_0) = f(v_0) - 1 = 0$ and $g(w) = f(w)$ for each $w \in V(W_n) - \{v_{1,k}, v_0\}$. Notice that $N(v_0) = \{v_{1,i}\} (1 \leq i \leq n)$. Since for each i , $\{v_{2,i}\} \subseteq N(v_{1,i}) \cap V_1'$, then f' is safe, f is a WRDF of W_n which implies $\gamma_w(W_n) \leq n+1$.

To see that $\gamma_w(W_n) \geq n+1$, suppose to the contrary that $\gamma_w(W_n) \leq n$. Let g be a WRDF of W_n with $w(g) = \gamma_w(W_n) \leq n$. By proposition 1, we must have $g(u_k) + g(v_{2,k}) = 1$ then v_0 becomes an undefended vertex which contradict to g is a WRDF of W_n . Therefore, $\gamma_w(W_n) = n+1$. $\square$

Theorem 4 Let G be a generalized web $W_{t,3}$ for $t \geq 3$, then the weak Roman domination number of G is $\gamma_w(G) = t+2$.

Proof. Define $f: V(G) \rightarrow \{0,1,2\}$ as $f(v_{t,k}) = 1$ for $1 \leq k \leq n$, $f(v_{i,1}) = 1$ for $1 \leq i \leq t-1$ and for other vertex w in G , $f(w) = 0$. Then $V_1 = \{v_{i,1}, v_{t,k}\}$ for $1 \leq i \leq t-1$, $1 \leq k \leq n$, $V_2 = \emptyset$ and $V_0 = V - V_1$ and $w(f) = t+2$. Since each vertex in V_0 is adjacent to a vertex in V_1 , f is a safe function with cost $t+2$. For each $u_k \in V_0$, consider a move f' from $v_{t,k}$ to u_k . Then, $V_0' = V_0 - u_k + v_{t,k}$, $V_1' = V_1 - v_{t,k} + u_k$ and $V_2' = \emptyset$. Notice that $N(v_{t,k}) = \{v_{t,k+1}, v_{t,k-1}, v_{t-1,k}, u_k\}$. Since $\{v_{t,k+1}, v_{t,k-1}, u_k\} \subseteq V_1'$, if $v_{t-1,k} \in V_1'$, then f' is safe; If $v_{t-1,k} \in V_0'$, then $N(v_{t-1,k}) \cap V_1' = \{v_{t-1,1}\}$ which implies f' is safe. For each $v_{i,2} \in V_0'$ for $2 \leq i \leq t-1$, consider a move g from $v_{i,1}$ to $v_{i,2}$. Notice that $N(v_{i,1}) = \{v_{i,2}, v_{i,3}, v_{i+1,1}, v_{i-1,1}\}$. Since $\{v_{i+1,1}, v_{i-1,1}, v_{i,2}\} \subseteq V_1'$ and $\{v_{i,2}\} \subseteq N(v_{i,3}) \cap V_1'$. Hence, g is safe. Similarly, there is a move from $v_{i,1}$ to $v_{i,3}$ which is safe. Since $v_{1,j}$ and v_0 forms a clique K_4 , and $v_{1,1} \in V_1$, each move between them is safe. Hence, f is a WRDF of G which implies $\gamma_w(G) \leq t+2$.

Next we show that $\gamma_w(G) \geq t+2$. Let f be an optimal WRDF of $W_{t,3}$. By proposition 1, $f(v_{3,i}) = 1$ for $1 \leq i \leq 3$. We claim that $\sum (f(v_{i,j})) \geq t-2$ for $2 \leq i \leq t-1$ and $1 \leq j \leq 3$. Since v_0 is at distance at least 2 from each vertex of C_3^i for $2 \leq i \leq t$, we have $\sum_{u \in N[v_0]} f(u) \geq 1$ where $N(v_0) + v_0$ is the closed neighborhood of v_0 . Then $w(f) \geq 3+t-2+1 = t+2$. Hence $\gamma_w(G) = t+2$. $\square$

Theorem 5 Let G be a generalized web $W_{3,4}$, then the weak Roman domination number of G is $\gamma_w(G) = 7$.

Proof. First we show that $\gamma_w(G) \leq 7$. Define

$f:V(G) \rightarrow \{0,1,2\}$ as $f(v_{3,k})=1$ for $1 \leq k \leq n$, $f(v_0)=1$, $f(v_{2,1})=1$ and $f(v_{1,3})=1$, for other vertex w in G , $f(w)=0$. Then, $V_1 = \{v_{3,k}, v_{2,1}, v_{1,3}, v_0\}$ for $1 \leq k \leq n$, $V_2 = \emptyset$ and $V_0 = V - V_1 - V_2$ and. Since each vertex in V_0 is adjacent to a vertex in V_1 , f is a safe function with $w(f) = 2|V_2| + |V_1| = 7$. By proposition 3, there is a safe move for each u_k . By proposition 6, there is a safe move from $v_{2,1}$ to $v_{1,1}$. For $v_{2,2}$, consider a move g_1 from $v_{2,1}$ to $v_{2,2}$. Then, $V_0' = V_0 - v_{2,2} + v_{2,1}$, $V_1' = V_1 - v_{2,1} + v_{2,2}$ and $V_2' = \emptyset$. Notice that $N(v_{2,1}) = \{v_{3,1}, v_{1,1}, v_{2,4}, v_{2,2}\}$. Since $\{v_{3,1}, v_{2,4}\} \subseteq V_1'$, $v_{2,4} \in N(v_{3,4})$ and $v_{1,1} \in N(v_0)$, then g_1 is safe. Similarly, there is a safe move from $v_{2,1}$ to $v_{2,4}$. For $v_{2,3}$, consider a move g_2 from $v_{1,3}$ to $v_{2,3}$. Then, $V_0' = V_0 - v_{2,3} + v_{1,3}$, $V_1' = V_1 - v_{1,3} + v_{2,3}$ and $V_2' = \emptyset$. Notice that $N(v_{1,3}) = \{v_0, v_{2,3}, v_{1,2}, v_{1,4}\}$. Since $\{v_0, v_{2,4}\} \subseteq V_1'$ and $\{v_{1,2}, v_{1,4}\} \subseteq N(v_0)$, then g_2 is safe. For $v_{1,2}$, consider a move g_3 from v_0 to $v_{1,2}$. Since $N(v_0) \subseteq N(v_{1,3}) \cup N(v_{2,1})$, then g_3 is safe. Hence, f is a WRDF of G which implies $\gamma_w(G) \leq 7$.

Next we show that $\gamma_w(G) \geq 7$. Let g be a WRDF of G . By proposition 1, $f(v_{3,k})=1$ ($1 \leq k \leq 4$). Let $|C_4^2 \cap (V_1 \cup V_2)| = t$. If $t < 1$, by proposition 3, we must have $f(v_{1,k}) \geq 1$ for $1 \leq k \leq 4$. Then, $w(g) \geq 4 + 4 = 8 > 6$. If $t > 1$, since v_0 is at distance at least two from $v_{2,i}$ and $v_{3,i}$ for $1 \leq i \leq 4$, $N[v_0] \cap (V_1 \cup V_2) \neq \emptyset$, we have $\sum_{u \in N[v_0]} f(u) \geq 1$. Hence, $w(g) \geq 4 + t + 1 \geq 7$. If $t = 1$ (say $v_{2,1}$), then we must have $f(v_{1,3}) = 1$. In order to avoid undefended vertex, we must have $f(v_0) = 1$ or $f(v_{1,1}) = 1$. Hence, $\gamma_w(G) \geq 4 + 1 + 2 = 7$. Hence, f is a WRDF of G with $\gamma_w(G) = 7$. $\square$

Theorem 6 Let G be a generalized web $W_{3,5}$, then the weak Roman domination number of G is $\gamma_w(G) = 9$.

Proof. First we show that $\gamma_w(G) \leq 9$. Define $f:V(G) \rightarrow \{0,1,2\}$ as $f(v_{3,k})=1$ for $1 \leq k \leq n$, $f(v_0)=2$, $f(v_{2,1})=1$ and $f(v_{2,4})=1$, for other vertex w in G , $f(w)=0$. Then, $V_1 = \{v_{3,k}, v_{2,1}, v_{2,3}\}$ for $1 \leq k \leq n$, $V_2 = \{v_0\}$ and $V_0 = V - V_1 - V_2$. Since each vertex in V_0 is adjacent to a vertex in V_1 , f is a safe function with $w(f) = 2|V_2| + |V_1| = 2 \cdot 1 + 7 = 9$. By proposition 3, there is a safe move for each u_k . By proposition 2, there is a safe move for each $v_{1,k}$ for $1 \leq k \leq 5$. For $v_{2,2}$, consider a move g from $v_{2,1}$ to $v_{2,2}$. Then, $V_0' = V_0 - v_{2,2} + v_{2,1}$, $V_1' = V_1 - v_{2,1} + v_{2,2}$ and $V_2' = V_2$. Since $N(v_{2,k}) \subseteq N(v_{3,k+1}) \cup N(v_{3,k-1}) \cup N(v_0)$ for all $1 \leq k \leq 5$, then g is safe. Similarly, there are safe moves from $v_{2,1}$ to $v_{2,5}$ and from $v_{2,4}$ to $v_{2,3}$. Hence, f is a WRDF of G which implies $\gamma_w(G) \leq 9$.

Next we show that $\gamma_w(G) \geq 9$. Let g be a WRDF of G . By proposition 1, $g(v_{3,k})=1$ ($1 \leq k \leq 5$). Let $|C_5^2 \cap (V_1 \cup V_2)| = t$. If $t < 2$, by proposition 3, there must be at least four vertices of C_5^1 in V_1 . Then, $w(g) \geq 5 + t + 4 \geq 9$. If $t > 2$, since v_0 is at distance at least two from $v_{2,i}$ and $v_{3,i}$ for $1 \leq i \leq 5$, $N[v_0] \cap (V_1 \cup V_2) \neq \emptyset$, we have $\sum_{u \in N[v_0]} f(u) \geq 1$. Hence, $w(g) \geq 5 + t + 1 \geq 9$. If $t = 2$, we know that $\sum_{u \in N[v_0]} f(u) \geq 1$, and we consider the following cases. If $f(v_0) = 0$, since there must exist a vertex $v_{1,k}$ for $1 \leq k \leq 5$ such that $(v_{1,k}, v_{1,j}) \notin E(W_{3,5})$ for $k \neq j$. we must have another two vertices in $V_1 \cup V_2$ station in C_n^1 , then $w(g) \geq 5 + 2 + 2 = 9$. If $f(v_0) = 1$, since there must exist a vertex $v_{1,k}$ for $1 \leq k \leq 5$ such that $(v_{1,k}, v_{1,j}) \notin E(W_{3,5})$ for $k \neq j$, consider a move to $v_{1,k}$, then $v_{1,j}$ becomes an undefended vertex which contradict to g is a

WRDF of $W_{3,5}$. Hence we have $\sum_{u \in N[v_0]} f(u) > 1$.

That is, $w(g) \geq 5+2+2=9$. Hence, $\gamma_w(G)=9$. $\square$

Theorem 7 Let G be a generalized web $W_{3,n}$ for $n \geq 6$. Let $n = 3a + b$ ($0 \leq b \leq 2$), then the weak Roman domination number of G is

$$\gamma_w(G) = \begin{cases} 4a+2 & \text{if } b=0; \\ 4a+3+b & \text{if } b>0. \end{cases}$$

Proof. According to the value of b , we have following cases.

Case 1: $b=0$. Define $f: V(G) \rightarrow \{0,1,2\}$ as $f(v_{3,k})=1$ for $1 \leq k \leq n$, $f(v_{2,3j+2})=1$ for $0 \leq j \leq a-1$ and $f(v_0)=2$, for other vertex w in G , $f(w)=0$. Then, $V_1 = \{v_{2,3j+2}, v_{3,k}\}$ for $0 \leq j \leq a-1$ and $1 \leq k \leq n$, $V_2 = \{v_0\}$ and $V_0 = V - V_1 - V_2$. Since $N(v_{1,k}) \cap V_2 = \{v_0\}$, $N(u_k) \cap V_1 = \{v_{3,k}\} \subseteq N(v_{2,k}) \cap V_1$ for $1 \leq k \leq n$, f is a safe function with $w(f) = 2|V_2| + |V_1| = 2 \cdot 1 + n + a = 4a + 2$. By proposition 1, there is a safe move for each u_k for $1 \leq k \leq n$. By proposition 2, there is a safe move for each $v_{1,k}$ for $1 \leq k \leq n$. For each $v_{2,3j+1} \in V_0$ for $0 \leq j \leq a-1$, consider a move h from $v_{2,3j+2}$ to $v_{2,3j+1}$. Notice that $N(v_{2,3j+2}) = \{v_{1,3j+2}, v_{3,3j+2}, v_{2,3j+1}, v_{2,3j+3}\}$. Since, $N(v_{1,3j+2}) \cap V_2' = \{v_0\}$, $\{v_{2,3j+1}, v_{3,3j+2}\} \subseteq V_1'$, and $N(v_{2,3j+3}) \cap V_1' = \{v_{3,3j+3}\}$, h is safe. Similarly, there is a safe move from $v_{2,3j+2}$ to $v_{2,3j+3}$. Hence, f is a WRDF of G which implies $\gamma_w(G) \leq 4a + 2$.

Next we show that $\gamma_w(G) \geq 4a + 2$. Let f be an optimal WRDF of $W_{3,n}$. By proposition 1, we have $f(v_{3,i})=1$ for $1 \leq i \leq n$. If $|C_n^2 \cap (V_1 \cup V_2)| = t < a$, then by proposition 3, there must be at least $3a - 3t$ vertices in $C_n^2 \cap V_0$ such that the safe move to those vertices is from C_n^1 . Then, $w(f) = 3a + t + 3a - 3t \geq 4a + 2$. when $t < a$. If $|C_n^2 \cap (V_1 \cap V_2)| > a$, since v_0 is at distance at least 2 from $v_{2,i}$ and $v_{3,i}$ for $1 \leq i \leq n$,

$N[v_0] \cap (V_1 \cup V_2) \neq \emptyset$. Therefore, $w(f) \geq 3a + a + 1 + 1 = 4a + 2$. If $|C_n^2 \cap (V_1 \cup V_2)| = a$, similarly, we have $\sum_{u \in N[v_0]} f(u) \geq 1$. If

$\sum_{u \in N[v_0]} f(u) = 1$, then we must have $f(v_0) = 1$ to avoid undefended vertex. Since there are at least $2a$ vertices in $C_n^1 \cap V_0$ without secured neighbor in C_n^2 , when there is a move from v_0 to one of those vertices, it creates at least $3a - (a + 3) = 2a - 3 > 0$ undefended vertex, which contradict to f is a WRDF of $W_{3,n}$. Hence, we have $\sum_{u \in N[v_0]} f(u) > 1$. That is, $w(f) \geq 3a + a + 2 = 4a + 2$. Hence, $\gamma_w(G) = 4a + 2$.

Case 2: $b > 0$. Define $f: V(G) \rightarrow \{0,1,2\}$ as $f(v_{3,k})=1$ ($1 \leq k \leq n$), $f(v_0)=2$, $f(v_{2,3j+1})=1$ for $0 \leq j \leq a-1$ and $f(v_{2,n-1})=1$, for other vertex w in G , $f(w)=0$. Then, $V_1 = \{v_{2,3j+1}, v_{3,k}, v_{2,n-1}\}$ for $0 \leq j \leq a-1$ and $1 \leq k \leq n$, $V_0 = V - V_1 - V_2$ and $V_2 = \{v_0\}$. Since each vertex in V_0 is adjacent to a vertex in V_1 or V_2 , f is a safe function with $w(f) = 2|V_2| + |V_1| = 2 \cdot 1 + n + a + 1 = 4a + b + 3$. For each unsecured vertex, there is a safe move similar to case 1. Hence, f is a WRDF of G which implies $\gamma_w(G) \leq 4a + 3 + b$.

Next we show that $\gamma_w(G) \geq 4a + 3 + b$. Let f be a optimal WRDF of $W_{3,n}$. By proposition 1, we have $f(v_{3,i})=1$ for $1 \leq i \leq n$.

Case 2.1: If $|C_n^2 \cap (V_1 \cup V_2)| > a + 1$, since v_0 is at distance at least two from $v_{2,i}$ and $v_{3,i}$ for $1 \leq i \leq n$, then $N[v_0] \cap (V_1 \cup V_2) \neq \emptyset$. Hence, $w(f) \geq 3a + b + a + 2 + 1 = 4a + 3 + b$.

Case 2.2: If $|C_n^2 \cap (V_1 \cup V_2)| = a + 1$, similarly, we have $\sum_{v \in N[v_0]} f(v) \geq 1$. If $\sum_{v \in N[v_0]} f(v) = 1$, then we must have $f(v_0) = 1$ to avoid undefended vertex. Since there are at least $2a + b - 1$ vertices in $C_n^1 \cap V_0$ without secured neighbor in C_n^2 , when there is a move from v_0 to one of those vertices, it creates at least $3a + b - (a + 1) - 3 = 2a + b - 4 > 0$

when $a \geq 2$ undefended vertex, which contradict to f is a WRDF of G . Hence, we have $\sum_{v \in N[v_0]} f(v) > 1$. That is, $w(f) \geq 3a + b + (a+1) + 2 = 4a + 3 + b$.

Case 2.3: If $|C_n^2 \cap (V_1 \cup V_2)| < a+1$, we have following cases.

Case 2.3.1: $|C_n^2 \cap (V_1 \cup V_2)| = a$. By proposition 3, there must be a vertex in $C_n^2 \cap V_0$ such that the safe move is from a vertex in C_n^1 , there are at least $3a+1-(a+1)=2a \geq 4$ vertices become undefended. We must have two more legions to station in C_n^1 which $w(f) = 3a+1+a+1+2 = 4a+4$, or we must have $f(v_0)=1$. If $f(v_0)=1$, since there at least $2a$ vertices in $C_n^1 \cap V_0$ without secured neighbor in C_n^2 , when there is a move from v_0 to those vertices, it creates at least $3a+1-a-3 \cdot 2 = 2a-5 > 0$ undefended vertices which contradict to f is a WRDF of G . Hence, $f(v_0)=2$. Then, $w(f) \geq 3a+1+1+2 = 4a+4$.

Case 2.3.2: $|C_n^2 \cap (V_1 \cup V_2)| \leq a-1$. By proposition 3, there must be at least $3a+b-(a-1) = 2a+b+1 > a+3$ vertices in $C_n^1 \cap (V_1 \cup V_2)$. Hence, $w(g) \geq 3a+b+a-1+2a+b+1 > 4a+3+b$. Hence, $\gamma_w(G) = 4a+3+b$ $\square$

Lemma 1 Let $t = 3c+d$ ($c \geq 1$). Figure 4 produced a WRDF of $W_{t,4}$ for $t \geq 4$ with cost $w(f) = 4(c+1)+d$.

Proof. According to the value of d , we have following cases.

Case 1: Let $d=1$. By Figure 4, the cost of the vertices in outer cycle is 4 (see line 2), and the center vertex cost is 1 (see line 3). For the cost of inner cycles, from line 6 to 8 with be repeated c times and each time with produce e vertices in V_1 , hence the total cost of $W_{t,n}$ is $w(f) = 4+1+4 \cdot c = 4(c+1)+1$. Since each vertex in V_0 is adjacent to a vertex in V_1 , f is safe. By proposition 1, there is a safe move for each u_k for $1 \leq k \leq n$. By proposition 5, there is a safe move of their neighbors for each $v_{3i+3,1}$

and $v_{3i+3,3}$ ($0 \leq i \leq c-1$). By proposition 6, there is a safe move for each $v_{3i+2,2}$ ($0 \leq i \leq c-1$) and $v_{3i+4,4}$ ($0 \leq i \leq c-2$). For $v_{2,4}$, consider a move g_1 from $v_{1,4}$ to $v_{2,4}$. Then $V_1' = V_1 - v_{1,4} + v_{2,4}$, $V_0' = V_0 - v_{2,4} + v_{1,4}$ and $V_2' = \emptyset$. Notice that $N(v_{1,4}) = \{v_{2,4}, v_0, v_{1,1}, v_{1,3}\}$. Since $\{v_{1,1}, v_{1,3}\} \subseteq N(v_0)$ and $\{v_{2,4}, v_0\} \subseteq V_1'$. Then g_1 is safe. For each k , $1 \leq k \leq 3$, consider a move g_2 from v_0 to $v_{1,k}$. Then $V_1' = V_1 - v_0 + v_{1,k}$, $V_0' = V_0 - v_{1,k} + v_0$ and $V_2' = \emptyset$. Since $\{v_{1,4}, v_{1,k}\} \in V_1'$ and $N(v_0) \subseteq N(v_{1,4}) \cup N(v_{1,k})$, g_2 is safe. Therefore, f is a WRDF of G .

Case 2: $d=2$. Similar to case 1, the cost of f produced by Figure 4 is $w(f) = 4+0+4 \cdot c + 2 = 4(c+1)+2$. Since each vertex in V_0 is adjacent to a vertex in V_1 , f is safe. By proposition 1, there is a safe move for each u_k for $1 \leq k \leq n$. By proposition 5, there is a safe move of their neighbors for each $v_{3i+4,1}$ and $v_{3i+4,3}$ ($0 \leq i \leq c-1$). By proposition 6, there is a safe move for each $v_{3i+3,2}$ and $v_{3i+2,4}$ ($0 \leq i \leq c-1$). For $v_{2,1}$, consider a move g_1 from $v_{1,1}$ to $v_{2,1}$. Then $V_1' = V_1 - v_{1,1} + v_{2,1}$, $V_0' = V_0 - v_{2,1} + v_{1,1}$ and $V_2' = \emptyset$. Notice that $N(v_{1,1}) = \{v_{1,4}, v_{1,2}, v_0, v_{2,1}\}$. Since $v_{2,1} \in V_1'$ and $\{v_{1,4}, v_{1,2}, v_0\} \subseteq N(v_{1,3})$, g_1 is safe. Similarly, there is a safe move from $v_{1,3}$ to $v_{2,3}$. For $v_{1,k} \in V_0$ for $k=2,4$ and v_0 , consider a move g_2 from $v_{1,1}$ to $v_{1,k}$ or v_0 . Since $v_{2,1} \in N(v_{2,4})$ and $N(v_{1,1}) - v_{2,1} \subseteq N(v_{1,3})$, g_2 is safe. Therefore, f is a WRDF of G .

Case 3: $d=0$. Similar to case 1, the cost of f produced by Figure 4 is $w(f) = 4+1+4 \cdot (c-1)+2+1 = 4(c+1)$. Since each vertex in V_0 is adjacent to a vertex in V_1 , f is safe. By proposition 1, there is a safe move for each u_k for $1 \leq k \leq n$. By proposition 5, there is a safe move of their neighbors for each $v_{3i+2,1}$ and $v_{3i+2,3}$ ($0 \leq i \leq c$). By proposition 6, there is a safe move for each $v_{3i+4,2}$ and $v_{3i+3,4}$ ($0 \leq i \leq c-1$). For

$v_{1,4}$, consider a move g_1 from v_0 to $v_{1,4}$. Then $V'_1 = V_1 - v_0 + v_{1,4}$, $V'_0 = V_0 - v_{1,4} + v_0$ and $V'_2 = \emptyset$. Notice that $N(v_0) = \{v_{1,k}\}$ for $1 \leq k \leq 4$. Since $\{v_{1,4}, v_{1,2}\} \subseteq V'_1$ and $N(v_0) \subseteq N(v_{1,2}) \cup N(v_{1,4})$, g_1 is safe. Therefore, f is a WRDF of G . $\square$

Algorithm 1 for $W_{t,4}$:

Input: Integers $t (t \geq 4)$ such that $t = 3c + d$ for $0 \leq d \leq 2$ represent the number of cycles in $W_{t,4}$;
 Output: A weak Roman domination function f of $W_{t,4}$;
 Method:
 1 Initialize $f(v) = 0$ for all $v \in W_{t,4}$.
 2 Let $f(v) = 1$ for all v in the outer cycle.
 3 If $d \neq 2$, let $f(v_0) = 1$.
 4 Let $h = t - 1$.
 5 Repeat
 6 $f(v_{h,1}) = 1; f(v_{h,3}) = 1; h = h - 1;$
 7 $f(v_{h,2}) = 1; h = h - 1;$
 8 $f(v_{h,4}) = 1; h = h - 1;$
 9 Until $h = 0$;
 End of Algorithm 1

Figure 4: Algorithm 1 for $W_{t,4}$.

Lemma 2 Let $t = 3c + d$ for $c \geq 1$. Figure 5 produced a WRDF of $W_{t,5}$ for $t \geq 4$ with cost

$$w(f) = \begin{cases} 5(c+1) + d + 1 & \text{if } d = 2; \\ 5(c+1) + d & \text{otherwise.} \end{cases}$$

Proof. According to value of d , we have following cases.

Case 1: $d = 1$. Similar to lemma 1, the cost of f produced by Figure 5 is $w(f) = 5 + 1 + 5 \cdot c = 5(c+1) + 1$. Since each vertex in V_0 is adjacent to a vertex in V_1 , f is safe. By proposition 1, there is a safe move for each u_k for $1 \leq k \leq n$. By proposition 5, there is a safe move of their neighbors for each $v_{t-1,1}$, $v_{3i+2,2}$, $v_{3i+1,3}$ and $v_{3i+3,4}$ ($0 \leq i \leq c-1$). By proposition 6, there is a safe move for each $v_{3i+3,1}$ ($0 \leq i \leq c-2$) and $v_{3i+2,5}$ ($0 \leq i \leq c-1$). For

$v_{1,1}$, consider a move g_1 from v_0 to $v_{1,1}$. Then $V'_1 = V_1 - v_0 + v_{1,1}$, $V'_0 = V_0 - v_{1,1} + v_0$ and $V'_2 = \emptyset$. Notice that $N(v_0) = \{v_{1,m}\}$ for $1 \leq m \leq 5$. Since $\{v_{1,1}, v_{1,3}\} \subseteq V'_1$, $\{v_{1,5}, v_{1,2}\} \subseteq N(v_{1,1})$ and $v_{1,4} \in N(v_{1,3})$, g_1 is safe. Therefore, f is a WRDF of G .

Case 2: $d = 2$. Similar to case 1, the cost of f produced by Figure 5 is $w(f) = 5 + 1 + 5 \cdot c + 2 = 5(c+1) + 3$. Since each vertex in V_0 is adjacent to a vertex in V_1 , f is safe. By proposition 1, there is a safe move for each u_k for $1 \leq k \leq n$. By proposition 5, there is a safe move of their neighbors for each $v_{t-1,1}$, $v_{3i+3,2}$, $v_{3i+2,3}$ and $v_{3i+3,4}$ ($0 \leq i \leq c-1$). By proposition 6, there is a safe move for each $v_{3i+4,1}$ ($0 \leq i \leq c-2$) and $v_{3i+2,5}$ ($0 \leq i \leq c-1$). For $v_{2,1}$ consider a move g_1 from $v_{1,1}$ to $v_{2,1}$. Then $V'_1 = V_1 - v_{1,1} + v_{2,1}$, $V'_0 = V_0 - v_{2,1} + v_{1,1}$ and $V'_2 = \emptyset$. Notice that $N(v_{1,1}) = \{v_0, v_{1,2}, v_{1,5}, v_{2,1}\}$. Since $\{v_0, v_{2,1}\} \subseteq V'_1$ and $\{v_{1,2}, v_{1,5}\} \subseteq N(v_0)$, g_1 is safe. For $v_{1,2}$, consider a move g_2 from v_0 to $v_{1,2}$. Since $N(v_0) \subseteq N(v_{1,1}) \cup N(v_{1,4})$, g_2 is safe. Similarly, there are safe moves from v_0 to $v_{1,3}$ and $v_{1,5}$. Therefore, f is a WRDF of G .

Case 3: $d = 0$. Similar to case 1, the cost of f produced by Figure 5 is $w(f) = 5 + 1 + 5 \cdot (c-1) + 2 + 2 = 5(c+1)$. Since each vertex in V_0 is adjacent to a vertex in V_1 , f is safe. By proposition 1, there is a safe move for each u_k for $1 \leq k \leq n$. By proposition 5, there is a safe move of their neighbors for each $v_{t-1,1}$, $v_{3i+3,3}$, $v_{3i+1,2}$ and $v_{3i+2,4}$ ($0 \leq i \leq c-1$). By proposition 6, there is a safe move for each $v_{3i+2,1}$ and $v_{3i+4,5}$ ($0 \leq i \leq c-2$). Therefore, f is a WRDF of G . $\square$

For $n = 5$, we set $f(v_0) = 1$ and modify algorithm 1 from line 6 to 8 as in Figure 4.

6	$f(v_{h,1})=1; f(v_{h,4})=1; h=h-1;$
7	$f(v_{h,2})=1; f(v_{h,5})=1; h=h-1;$
8	$f(v_{h,3})=1; h=h-1;$

Figure 5: Algorithm 2 for $W_{t,n}$, modify from Algorithm 1.

Lemma 3 Let $n=3a(a \geq 2)$. Figure 6 produced a WRDF of $W_{t,n}$ for $t \geq 4$ and $n \geq 6$ with cost $w(f) = n(c+1) + 2 - (2-d)a$.

Proof. Similar to lemma 1, the cost of f produced by Figure 6 is

$$\begin{aligned} w(f) &= 3a + (t-2)a + 2 \\ &= n + 2 - (2-d)a. \end{aligned}$$

Since each vertex in V_0 is adjacent to a vertex in V_1 , f is safe. By proposition 1, there is a safe move for each u_k for $1 \leq k \leq n$. By proposition 2, there is a safe move of their neighbors for each v_0 . By proposition 5, there is a safe move from other vertices in V_1 to each of their neighbors. Therefore, f is a WRDF of G . $\square$

For $n=3a(a \geq 2)$, we set $f(v_0)=2$ and modify algorithm 1 from line 5 to 9 as in Figure 5.

5 Repeat
6 for $i=0$ to $a-1$
7 $f(v_{h,3i+1})=1;$
8 $h=h-1;$
9 for $i=0$ to $a-1$
10 $f(v_{h,3i+2})=1;$
11 $h=h-1;$
12 for $i=0$ to $a-1$
13 $f(v_{h,3i+3})=1;$
14 $h=h-1;$
15 Until $h=1;$

Figure 6: Algorithm 3 for $W_{t,n}$, modify from Algorithm 1.

Lemma 4 Figure 7 produced a WRDF of $W_{t,n}$, where $t=3c+d$ for $0 \leq d \leq 2$ and $n=3a+1$ for $a \geq 2$ with cost $w(f) = n(c+1) + 3 - (2-d)a$.

Proof: We prove lemma according to value d with the following cases.

Case 1: $d=1$. Similar to lemma 1, the cost of f produced by Figure 7 is

$$\begin{aligned} w(f) &= 3a+1+2+(a+1+a+a)(c-1)+2a+1+1 \\ &= n(c+1)+3-a. \end{aligned}$$

Since each vertex in V_0 is adjacent to a vertex in V_1 , f is safe. By proposition 1, there is a safe move for each u_k for $1 \leq k \leq n$. By proposition 2, there is a safe move of their neighbors for each v_0 . By proposition 6, there is a safe move for each $v_{3i+2,n-2}$ ($0 \leq i \leq c-1$) and $v_{3i+4,n}$ ($0 \leq i \leq c-2$). By proposition 5, there is a safe move from other vertices in V_1 to each of their neighbors. Therefore, f is a WRDF of G .

Case 2: $d=2$. Similar to case 1, the cost of f produced by Figure 7 is

$$\begin{aligned} w(f) &= 3a+1+2+(a+1+a+a)c+1 \\ &= n(c+1)+3. \end{aligned}$$

Since each vertex in V_0 is adjacent to a vertex in V_1 , f is safe. By proposition 1, there is a safe move for each u_k for $1 \leq k \leq n$. By proposition 2, there is a safe move of their neighbors for each v_0 and $v_{2,n}$. By proposition 6, there is a safe move for each $v_{3i+3,n-2}$ and $v_{3i+5,n}$ ($0 \leq i \leq c-1$). By proposition 5, there is a safe move from other vertices in V_1 to each of their neighbors. Therefore, f is a WRDF of G .

Case 3: $d=0$. Similar to case 1, the cost of f produced by Figure 7 is

$$\begin{aligned} w(f) &= 3a+1+2+(a+1+a+a)(c-1)+a+1+1 \\ &= n(c+1)+3-2a. \end{aligned}$$

Since each vertex in V_0 is adjacent to a vertex in V_1 , f is safe. By proposition 1, there is a safe move for each u_k for $1 \leq k \leq n$. By proposition 2, there is a safe move of their neighbors for each v_0 and $v_{2,1}$. By proposition 5, there is a safe move of their neighbors for each $v_{3i+5,1}$, $v_{3i+5,n-1}$, $v_{3i+4,3j+2}$, $v_{3i+3,3j+3}$ ($0 \leq i \leq c-2$ and $0 \leq j \leq a-2$) and $v_{3i+2,3j+4}$ ($0 \leq i \leq c-1$ and $0 \leq j \leq a-2$). By proposition 6, there is a safe move for each $v_{3i+3,n}$ and $v_{3i+4,n-2}$ ($0 \leq i \leq c-2$). For $v_{3,n-1}$, consider a move g_1 from $v_{2,n-1}$ to $v_{3,n-1}$. Then $V'_1 = V_1 - v_{2,n-1} + v_{3,n-1}$, $V'_0 = V_0 - v_{3,n-1} + v_{2,n-1}$ and $V'_2 = V_2$. Notice that $N(v_{2,n-1}) =$

$\{v_{1,n-1}, v_{3,n-1}, v_{2,n}, v_{2,n-2}\}$. Since $v_{3,n-1} \in V_1'$, $\{v_{1,n-1}\} \subseteq N(v_0)$, $\{v_{2,n-2}\} \subseteq N(v_{2,n-3})$ and $\{v_{2,n}\} \subseteq N(v_{2,1})$, g_1 is safe. Therefore, f is a WRDF of G . $\square$

For $n = 3a+1$ ($a \geq 2$), we set $f(v_0) = 2$, modify algorithm 1 from line 6 to 14 as in Figure 6.

6	for $i=0$ to $a-1$
7	$f(v_{h,3i+1}) = 1;$
8	$f(v_{h,n-1}) = 1;$
9	$h = h-1;$
10	for $i=0$ to $a-1$
11	$f(v_{h,3i+2}) = 1;$
12	$h = h-1;$
13	for $i=0$ to $a-2$
14	$f(v_{h,3i+3}) = 1;$
15	$f(v_{h,n}) = 1;$
16	$h = h-1;$
18	if $d = 0$
19	$f(v_{2,1}) = 2;$
20	else if $d = 1$
21	$f(v_{2,n}) = 1;$
22	else $d = 2$
23	$f(v_{2,n}) = 2;$

Figure 7: Algorithm 4 for $W_{t,n}$, modify from Algorithm 1.

Lemma 5 Figure 8 produced a WRDF of $W_{t,n}$, where $t = 3c+d$ for $0 \leq d \leq 2$ and $n = 3a+2$ ($a \geq 2$) with cost $w(f) =$

$$\begin{cases} n(c+1)+2-(2-d)a & \text{if } d=0 \text{ or } (d=1, c=1); \\ n(c+1)+3-(2-d)a & \text{otherwise;} \end{cases}$$

Proof: We prove lemma according to value d with the following cases.

Case 1: $d=1$. We prove case 1 according to value c with the following cases.

Case 1.1: $c=1$. Similar to lemma 1, the cost of f produced by Figure 8 is

$$w(f) = 3a+2+2+(a+1+a+1+a)(c-1)+2a+2 = n(c+1)+2-a.$$

Since each vertex in V_0 is adjacent to a vertex in V_1 , f is safe. By proposition 1, there is a safe

move for each u_k for $1 \leq k \leq n$. By proposition 2, there is a safe move of their neighbors for each v_0 . By proposition 5, there is a safe move from other vertices in V_1 to each of their neighbors. Therefore, f is a WRDF of G .

Case 1.2: $c=1$. Similar to case 1.1, the cost of f produced by Figure 8 is

$$w(f) = 3a+2+2+(a+1+a+1+a)(c-1)+2a+3 = n(c+1)+3-a.$$

Since each vertex in V_0 is adjacent to a vertex in V_1 , f is safe. By proposition 1, there is a safe move for each u_k for $1 \leq k \leq n$. By proposition 2, there is a safe move of their neighbors for each v_0 and $v_{2,2}$. By proposition 6, there is a safe move for each $v_{3i+2,n}$ ($0 \leq i \leq c-1$) and $v_{3i+3,1}$ ($0 \leq i \leq c-2$). By proposition 5, there is a safe move from other vertices in V_1 to each of their neighbors. Therefore, f is a WRDF of G .

Case 2: Similar to case 1, the cost of f produced by Figure 8 is

$$w(f) = 3a+2+2+(a+1+a+1+a)c+1 = n(c+1)+3.$$

Since each vertex in V_0 is adjacent to a vertex in V_1 , f is safe. By proposition 1, there is a safe move for each u_k for $1 \leq k \leq n$. By proposition 2, there is a safe move of their neighbors for each v_0 . By proposition 5, there is a safe move from other vertices in V_1 to each of their neighbors. Therefore, f is a WRDF of G .

Case 3: Similar to case 1, the cost of f produced by Figure 8 is

$$w(f) = 3a+2+2+(a+1+a+1+a)(c-1)+a+1+1 = n(c+1)+2-2a.$$

Since each vertex in V_0 is adjacent to a vertex in V_1 , f is safe. By proposition 1, there is a safe move for each u_k for $1 \leq k \leq n$. By proposition 2, there is a safe move of their neighbors for each v_0 and $v_{2,1}$. By proposition 6, there is a safe move for each $v_{3i+4,n}$ ($0 \leq i \leq c-2$). By proposition 5, there is a safe move from other vertices in V_1 to each of their neighbors. Therefore, f is a WRDF of G . $\square$

For $n = 3a + 2$ ($a \geq 2$), we set $f(v_0) = 2$, and modify algorithm 4 from line 6 to 16 and line 20 to 23 as in Figure 7.

```

6      for  $i = 0$  to  $a$ 
7           $f(v_{h,3i+1}) = 1$ ;
8       $h = h - 1$ ;
9      for  $i = 0$  to  $a$ 
10          $f(v_{h,3i+2}) = 1$ ;
11      $h = h - 1$ ;
12     for  $i = 0$  to  $a - 1$ 
13          $f(v_{h,3i+3}) = 1$ ;
14      $h = h - 1$ ;
18 if  $d = 0$ 
19      $f(v_{2,1}) = 2$ 
20 else if  $d = 1$  and  $c \geq 2$ 
21      $f(v_{2,2}) = 2$ ;
22 else  $d = 2$ 
23      $f(v_{2,1}) = 1$ ;

```

Figure 8: Algorithm 5 for $W_{t,n}$, modify from Algorithm 1.

4 Conclusion

Weak Roman domination may be used to solve resources sharing problems. Take projector for an example. A projector is needed for group discussion. Since the budget is limited, not every lab has their own projector. However, the labs are closed to each other. We may use graph G to represent the distance relation between labs where each vertex represents a lab and if two labs are closed to each other, there is an edge between them. Each lab may have zero, one or two projectors depending on the budget of the lab. In order to have every lab be able to use a projector when it need, a lab without a projector must be adjacent to a lab having one or two projectors. If a lab needs a projector, one of its closed labs with one or two projectors may dispatch a projector to it. Therefore, we can solve this problem by weak Roman domination and find the minimum cost of the graph G is the same as buying minimum number of projectors such that every lab will be able to use one when it needs. In this paper, we apply the weak Roman domination on corona graphs and some generalized web graphs to find the minimum cost by provided algorithms.

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